Semestral

Elementary Number Theory

Instructor: Ramdin Mawia | Marks: **50** | **Time:** November 14, 2025; 14:00–17:00.

Attempt FIVE problems. The maximum you can score is 50.

1. Prove or disprove (**any two**): **5+5=10**
   i. For all positive integers a and n, the congruence $a^n \equiv a^{n-\varphi(n)} \pmod{n}$ holds.
   ii. There are integers a and b such that $a^2 + 2025a + 2027b = 2035$.
   iii. For any integer $n > 1$, the integer $n^4 + 1$ is not a power of 2 and any odd prime divisor of it is of the form $8k + 1$ for some $k \in \mathbb{N}$.

2. Show that $(x^2 - 3)/(3y^2 + 5)$ is never an integer for $x, y \in \mathbb{Z}$. **10**

3. Prove that $\alpha \in \mathbb{R}$ is irrational if and only if there are infinitely many distinct rational numbers a/b (with $\gcd(a, b) = 1$ and $b > 0$) such that $|\alpha - a/b| < 1/b^2$. **10**

4. Determine all primes p which are expressible in the form $p = x^2 + xy + y^2$ for some $x, y \in \mathbb{Z}$. [*Hint.* Think of class numbers!] **10**

OR

4.′ Prove that there are infinitely many primes of the form $8k - 1$. [*Hint.* Let $p_1, \ldots, p_n$ be primes of the form $8k - 1$. Look at $(p_1 \cdots p_n)^2 - 2$.]

5. Find all reduced binary quadratic forms (a, b, c) such that $a > 0$ and $b^2 - 4ac = 53$. **10**

6. Let μ be the Möbius function and let $\mu_2 = \mu * \mu$. Describe the set $\{\mu_2(n) : n \geqslant 1\}$. **10**

OR

6.′ Prove that

$$\sum_{n \leqslant x} \mu(n) \left[\sqrt{x/n}\right] = \sum_{n \leqslant x} (-1)^{\Omega(n)}$$

for all $x \geqslant 1$, where $\Omega(n)$ is the total number of prime factors of n, counted with multiplicity, i.e., if $n = p_1^{e_1} \cdots p_k^{e_k}$ is the standard factorisation of n, then $\Omega(n) = e_1 + \cdots + e_k$.

7. Let q be a fixed positive integer. Evaluate the sum **10**

$$\sum_{\substack{n \leqslant x \\ \gcd(n,q)=1}} \frac{1}{n}$$

with an error term which tends to 0 as $x \to \infty$.

8. Let $(u_n)_{n \geqslant 0}$ be an integer sequence satisfying a linear recurrence of order 2. Suppose $u_0 = 0, u_1 = 1, u_2 = 3, u_3 = 10$. Prove that if p is an odd prime with $(13/p) = -1$ (i.e., 13 is a quadratic nonresidue mod p), then $p | u_{p+1}$. **10**

The End